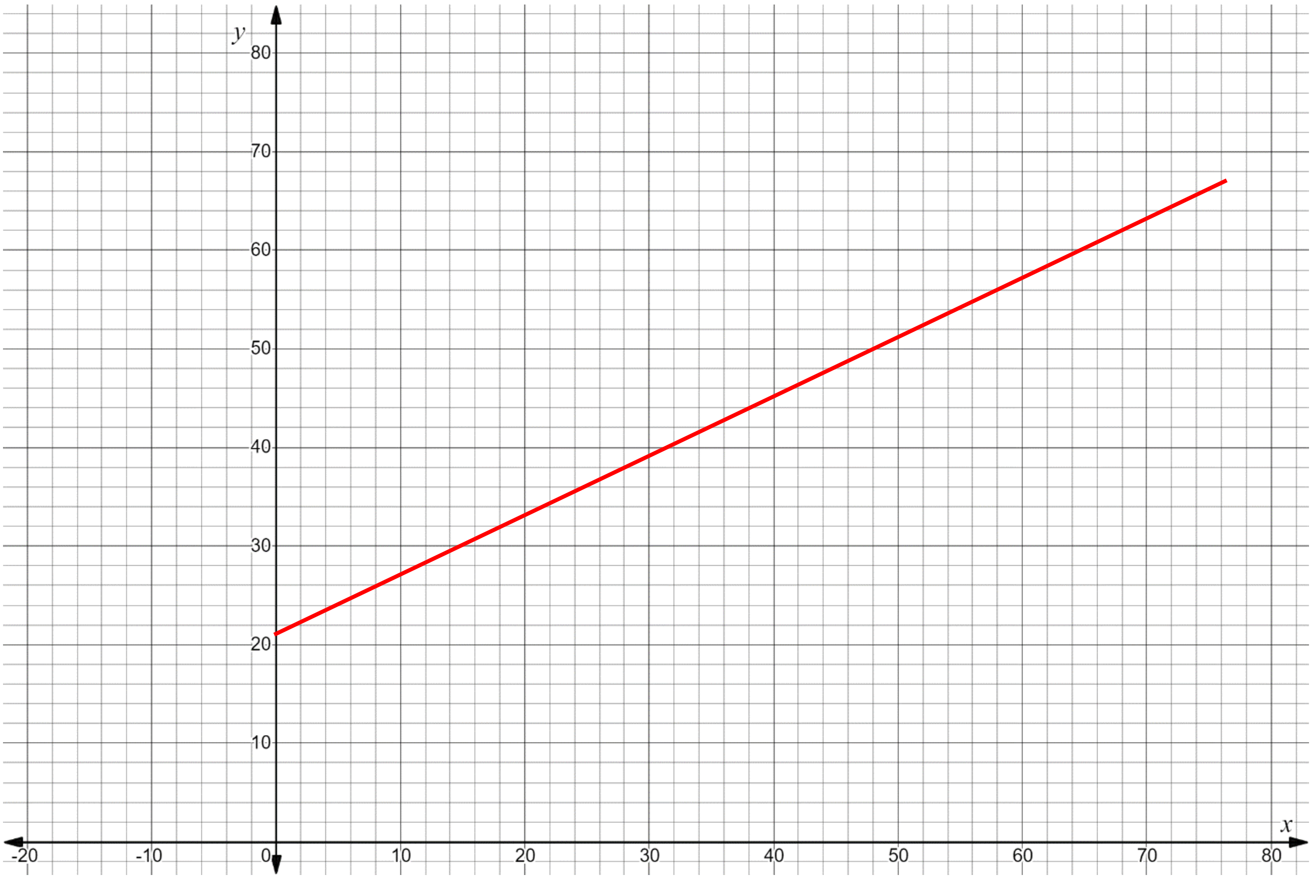
Algebra 1 | Name Answer Key___________
Unit 4 | Date____________________
Lesson 5 Homework | Period___________________

The table of values shows the linear relationship between x, the number of years since 1900 for all years between 1900 and 2000, and y, the percentage of people in the world who can read.

x	**Number of years since 1900**	0	10	32	48	73
y	**Percentage of people in the world who can read**	21	27	40.2	49.8	64.8

1. Graph the linear relationship.



2. Place a check in the box of a key feature that you used to help you graph the relationship.

Domain	**Range**	**x-intercept**	**y-intercept**	**Slope**	**Increasing / Decreasing**
✓	✓		✓	✓	✓

3. The equation $-6x + 10y = 210$ models the linear relationship. Place a check in the box of a key feature that can be determined using the equation.

Domain	**Range**	**x-intercept**	**y-intercept**	**Slope**	**Increasing / Decreasing**
✓	✓	✓	✓	✓	✓

4. Write the equation $-6x + 10y = 210$ in slope-intercept form.

$\frac{10y}{10} = \frac{6x}{10} + \frac{210}{10}$ $\boxed{y = \frac{3}{5}x + 21}$

5. What domain makes sense for the context?

$\{x | 0 \leq x \leq 100\}$

6. What range makes sense for the context?

$\frac{3}{5}\left(\frac{\overset{20}{\cancel{100}}}{1}\right) + 21$ $\boxed{\{y | 21 \leq y \leq 81\}}$

$60 + 21 = 81$

7. Determine the x –intercept.

$0 = \frac{3}{5}x + 21$

$\frac{5}{\underset{1}{\cancel{3}}} \cdot \frac{\overset{-7}{\cancel{-21}}}{1} = \frac{\cancel{5}}{\cancel{3}} \frac{\cancel{3}}{\cancel{5}} x$

$-35 = x$ $\boxed{(-35, 0)}$

8. Interpret the x –intercept.

35 years before 1900, 0% of the people in the world could read. Not within constraints of this scenario.

9. Determine the slope.

$\frac{3}{5}$

10. Interpret the slope.

There is a 3% increase of the people in the world who can read every 5 years.